



# Instructor Solution

Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Spring 2026

Instructor – Dr. Aamir Hasan - Section L4

Time = 20 Minutes

Quiz No 02

[20 points]

Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.

## Question 1 [10 points] [Bayes Rule / Conditional Probability]

CLO Assessed – CLO 05, Analyse a random experiment for presence of dependent situations and modelling and analysing it using notions of conditional probability - (Cog - 4).

Suppose 10 in 1000 people carry a certain disease (X), for which there is a pretty reliable test. Probability of a false negative (carrier tests negative) is 2% (so probability of a carrier of disease tests positive is 98%). Probability of a false positive (a non-carrier tests positive) is 10%. If a person just tested positive, what are the chances s/he is a carrier of the disease?

$D \rightarrow$  Person carrying Disease,  $D^c \rightarrow$  Person Not carrying disease

$$P(D) = 0.01$$

$$P(D^c) = 0.99$$

$T \rightarrow$  Test results positive

$T^c \rightarrow$  Test results Negative

$$P(T|D) = 0.98$$

$$P(T^c|D) = 0.02$$

$$P(T^c|D^c) = 0.9$$

$$P(T|D^c) = 0.1$$

$$P(D|T) = \frac{P(T|D) \cdot P(D)}{P(T)}$$

$$P(T) = P(T|D) \cdot P(D) + P(T|D^c) \cdot P(D^c)$$

$$= \frac{0.98 \times 0.01}{0.98 \times 0.01 + 0.1 \times 0.99}$$

$$= \frac{0.0098}{0.0098 + 0.099} = \frac{0.0098}{0.1088}$$

$$= 0.090$$

$$= 9\%$$

**Question 2 [10 points] [Conditional Probability]**

CLO Assessed – CLO 05, Analyse a random experiment for presence of dependent situations and modelling and analysing it using notions of conditional probability - (Cog - 4).

A hiker starts by taking one of 3 available trails, denoted as 1, 2, and 3. An hour into the hike (assuming normal speed), each trail  $i$  (where  $i = 1, 2, 3$ ) subdivides into  $i+1$  sub-trails. For each trail, only one of the sub-trails leads to the hiker's destination. The hiker has no map and makes random choices (uniform distribution) of both the trail and sub-trails.

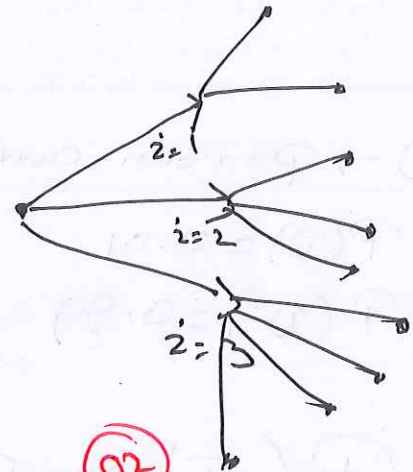
**Part a [5 points]** What is the probability that the hiker reaches the destination?

**Part b [5 points]** Given that the hiker reached the destination, what is the probability that the hiker took trail No 1?

(a)  $P[\text{hiker Reaches Destination}]$   
 $D = \text{hiker Reaches destination}$

$$P(D) = P(D|\text{Trail 1}) \cdot P(\text{Trail 1}) + P(D|\text{Trail 2}) \cdot P(\text{Trail 2}) + P(D|\text{Trail 3}) \cdot P(\text{Trail 3}).$$

$$= \frac{1}{2} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{1}{4}$$
$$= 0.361$$



$$(b) P(\text{Trail 1} | D) = \frac{P(D|\text{Trail 1}) \cdot P(\text{Trail 1})}{P(D)}$$

$$= \frac{\frac{1}{2} \cdot \frac{1}{3}}{0.361} = 0.461$$